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Elen-t: A Genetic Programming Engine for Symbolic Discovery in High-Dimensional Spaces

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Software

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Summary

Elen-t is an open-source Python library for symbolic regression: it recovers interpretable mathematical expressions from high-dimensional datasets using genetic programming. The ecosystem includes Elen-k, a unified dashboard for managing computational experiments.

Statement of need

In fields such as physics, chemistry, and drug discovery, researchers require not only accurate predictions but also the underlying mathematical laws. Black-box models (e.g., Random Forest, XGBoost) fail in high-dimensional spaces. Elen-t addresses this by providing a robust engine that actively constructs mathematical expressions.

State of the field

Current symbolic regression tools (e.g., gplearn, PySR) are powerful but often lack an integrated environment for managing high-dimensional benchmarks. Our benchmarks demonstrate that while standard models (RF, XGBoost) and baseline GP (gplearn) fail catastrophically in 128-dimensional spaces (achieving R^2 scores near zero or negative), Elen-t successfully recovers the underlying formulas, achieving R^2 scores up to 0.995.

Software design

The core module implements a strongly-typed genetic-programming loop. To handle high-dimensional spaces, we introduce a *log-grouped aggregation* frontend. For a 1024-dimensional raw input, features are aggregated into 4 log-transformed groups before symbolic regression. NumPy and Pandas handle data I/O; Elen-k is a React frontend.

Research impact statement

Elen-t has demonstrated significant potential in drug discovery, recovering nonlinear molecular relationships with 99.5% accuracy while producing human-readable formulas.

AI usage disclosure

AI-assisted tools were used for code scaffolding and documentation editing. All algorithms, scientific logic, and validation were designed and verified by the author.

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References

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